AI 时代商业工程 · 课题研究系列
BUSINESS ENGINEERING IN THE AI ERA · RESEARCH STUDY SERIES

No. 04 / 12

课题研究报告 · RESEARCH STUDY

AI经济学：为工作定价、构建毛利与重读资本账本

AI Economics: Pricing the Work, Building the Margin and Rereading the Capital Ledger

关于单位经济、定价工具与基础设施融资的课题研究

A research study on unit economics, pricing instruments and the financing of the AI build

研究底本 SOURCE
Cuofano, G. AI Economics — The Discipline of the Money in the AI Era (2026)

版本 EDITION
2026 年 10 月 · October 2026

语言 LANGUAGE
中英双语 · Bilingual Chinese–English

摘要 *Abstract*

本研究以Cuofano（2026）《AI Economics》为底本，考察人工智能时代的“记账方式”如何被重写。软件时代的财务建立在复制品边际成本为零的偶然条件之上；AI时代出售的是有成本的工作单元，由此产生三重倒转：**边际成本回归、席位从两端断裂、资本迁出资产负债表**。研究构建“单位—企业—资本”三层框架，以底本的机制、公开案例及2026年第三季度读数为证据。研究发现：第一，单位接受成果成本是唯一耐久的计量单位，毛利由路由、缓存、自有模型与注意力四个阀门设计而成；第二，重尾消费与智能体使席位定价失效，定价应沿可归因性与自主性网格迁移；第三，资本应以收入对折旧、回报门槛与融资性质衡量——AI收入约为折旧的一倍，维持覆盖需年增约44%，行业处于约6000亿美元门槛的三分之一，增量债务占资本开支比重在八个季度内由不足一成升至近三分之一。融资结构是放大器，而非泡沫本身。

This study takes Cuofano (2026), *AI Economics*, as its source and examines how the AI era rewrites the way the books are kept. Software-era finance rested on an accident, the zero marginal cost of a copy; the AI era sells units of work that cost something to produce, yielding three inversions: **marginal cost returns**, **the seat breaks from both ends**, and **capital moves off the balance sheet**. The study builds a three-level framework of unit, firm and capital, using the source's mechanisms, public cases and Q3 2026 readings as evidence. It finds, first, that cost per accepted outcome is the only durable unit of measurement and that gross margin is designed through four valves: routing, caching, an owned model and attention. Second, heavy-tailed consumption and agents break seat pricing; pricing should migrate along a grid of attribution and autonomy. Third, capital should be judged by revenue against depreciation, the return hurdle and the character of funding: AI revenue covers depreciation about once, holding that coverage needs roughly 44 percent annual growth, the industry stands at about a third of a $600 billion hurdle, and incremental debt rose from under a tenth to nearly a third of capex in eight quarters. The financing is an amplifier, not the bubble.

关键词 AI经济学；单位接受成果成本；毛利设计；定价网格；折旧覆盖；酸性测试

KEYWORDS AI economics; cost per accepted outcome; designed margin; pricing grid; depreciation coverage; acid test

研究问题 *Research Questions*

RQ1	边际成本回归后，什么单位能耐久地衡量AI产品的单位经济，毛利由什么决定?	Once marginal cost returns, what unit durably measures AI unit economics, and what sets gross margin?
RQ2	面对重尾消费与智能体，定价与增长核算应如何重建?	Facing heavy tails and agents, how should pricing and growth accounting be rebuilt?
RQ3	应以何种检验判断AI基础设施资本能否得到回报?	What tests should judge whether AI infrastructure capital is being repaid?
RQ4	融资结构的转变在多大程度上构成系统性风险，风险集中于何处?	How far does the financing shift create systemic risk, and where does it concentrate?

目录 *Contents*

01 研究背景与问题提出
Background and Problem Statement 4

02 核心概念与理论框架
Core Concepts and Analytical Framework 6

03 研究设计与方法
Research Design and Method 8

04 发现一：边际成本回归与毛利的设计化
Finding I: The Return of Marginal Cost and the Designed Margin 10

05 发现二：席位的断裂与定价网格
Finding II: The Broken Seat and the Pricing Grid 13

06 发现三：资本从配置变为义务
Finding III: Capital Turns from Allocation into Obligation 15

07 案例研究与比较分析
Case Studies and Comparative Analysis 19

08 讨论：启示、局限与风险
Discussion: Implications, Limitations and Risks 22

09 结论与未来研究方向
Conclusions and Future Research Agenda 24

— 参考文献与研究说明
References & Research Notes 26

第 1 章 · CHAPTER 1

01 研究背景与问题提出

Background and Problem Statement

软件时代的财务建立在“复制品零成本”这一偶然条件之上；人工智能时代撤走了这一条件，财务报表的变化顺序随之倒转。

中文

Software-era finance rested on an accident, the copy that cost nothing; the AI era removed it, and the order in which the financial statements change was reversed.

ENGLISH

1.1 从“网络平方”说起

Starting from Web Squared

人工智能并非在空地上兴起。它建立在三十年互联网积累的基础设施、数据、用户行为与分发渠道之上：网络提供了训练语料，也提供了触达十亿用户的通道。底本将这种关系称为**网络平方**（Web Squared）——模型不是取代网络，而是对网络进行复利放大。因此，首批被改造的行业恰恰是网络原生行业：搜索、社交、出版、软件与云。需求无需重新组装，因为网络早已把它组装好了。

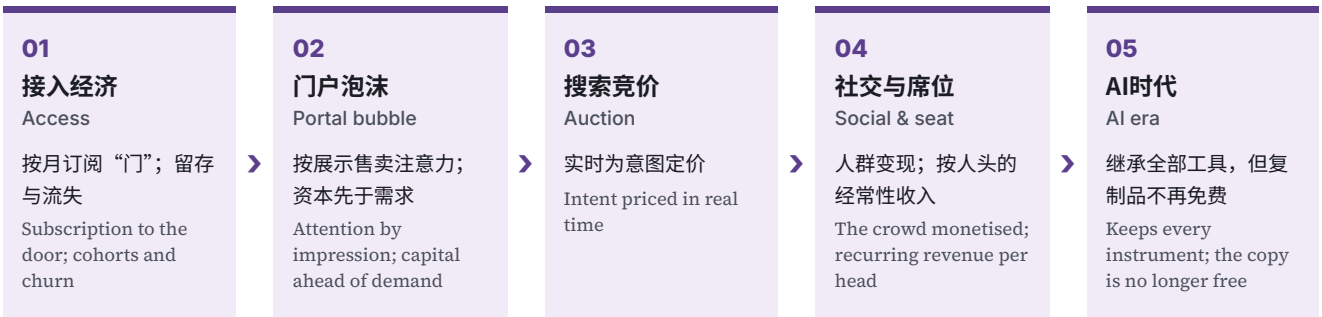
AI did not arise on empty ground. It sits on thirty years of web infrastructure, data, behaviour and distribution: the web supplied the training corpus and the channels that reach a billion users. The source volume calls this relationship **Web Squared**: the models compound the web rather than replace it. The first industries transformed are therefore the web-native ones, namely search, social, publishing, software and cloud. Demand did not have to be assembled, because the web had already assembled it.

但资金的流动方式这一次截然不同。网络经济建立在服务边际成本近乎为零的复制品之上，而人工智能经济建立在每生产一次都要付出成本的“工作单元”之上。这一事实使报表变化的顺序发生倒转：成本线先于收入线变动；被软件时代遗忘的物理约束——销货成本、折旧、先于需求投入的资本——重新回到了技术栈的顶端。

Yet the money moves differently this time. The web's economics were built on a copy that cost almost nothing to serve; the AI era's are built on a unit of work that costs something every time it is produced. That fact reverses the order in which the statements change: the cost line moves before the revenue line, and the physics the software era had forgotten, namely cost of goods, depreciation and capital committed ahead of demand, returns to the top of the stack.

图 1-1 网络经济的五种商业模式及其留给AI时代的财务工具

Figure 1-1 *Five web business models and the instruments each left to the AI era*



1.2 历史的韵脚：稀缺投入与资本周期
Historical rhymes: scarce inputs and capital cycles

底本对网络史的重读提供了两条与当下高度押韵的经验。其一来自接入经济：拥有“门”的企业只有在接入稀缺时才能收取租金，一旦稀缺投入变得充裕，租金便转移到该投入原本所把守的东西上。其二来自电信与互联网泡沫：资本按物理时钟先于需求投入，损益表被要求在到期日之前作答，需求最终如期而至，但回报落入了在破产后低价购得资产的人手中。

The source's rereading of web history yields two lessons that rhyme closely with the present. The first comes from the access economy: the firm that owns the door earns a rent only while access is scarce, and once the scarce input becomes abundant the rent moves to whatever the input was gating. The second comes from the telecom and dot-com cycles: capital committed on a physical clock ahead of demand, an income statement asked to answer before the maturity schedule, and demand that did arrive, with the returns going to whoever bought the assets out of bankruptcy.

1亿
两个月内的用户数
users in two months

~90%
单位智能价格年降幅
fall in unit price of intelligence, one year

~80%
软件时代典型毛利率
typical software-era gross margin

来源 Source: Cuofano (2026); consumer chat adoption (late 2022) and price curve as reported in the volume.

1.3 三重倒转与问题提出
Three inversions and the research problem

底本将时代变化归纳为三重倒转。第一，**边际成本回归**：软件出售复制品，而这个时代出售工作，工作有销货成本，即按词元计价、价格曲线不受企业控制的推理成本。第二，**席位从两端断裂**：消费呈重尾分布，而智能体完成的工作背后并无可计费的人头。第三，**资产负债表发生迁移**：生产智能的资本提前数年按物理时钟投入，且越来越多地通过供应商、租赁与私募信贷在表外融资。

The source condenses the shift into three inversions. First, **marginal cost is back**: software sold copies, this era sells work, and work carries a cost of goods, namely inference priced per token on a curve the firm does not control. Second, **the seat broke from both ends**: consumption is heavy-tailed, and agents complete work with no person behind it to charge. Third, **the balance sheet moved**: the capital that produces intelligence is committed years ahead on a physical clock and financed increasingly off balance sheet, through suppliers, leases and private credit.

由此，本研究的核心问题是：当软件财务的底层假设失效后，企业应当用什么样的计量单位、定价工具与资本检验来重新“记账”？本研究并不预测市场走向，而是检验底本提出的一套财务工具能否在单位、企业与产业三个层面上一致地解释2026年第三季度可观察到的现象。

The core problem of this study follows: once the founding assumption of software finance fails, what unit of measurement, pricing instrument and capital test should a firm use to keep its books again? The study does not forecast markets. It tests whether the set of finance instruments proposed by the source can explain, consistently at the levels of the unit, the firm and the industry, what was observable in the third quarter of 2026.

> **为工作定价，构建毛利，掌握账本。**
>
> *Price the work. Build the margin. Own the ledger.*
>
> — Cuofano, *AI Economics* (2026)

第 2 章 · CHAPTER 2

02 核心概念与理论框架

Core Concepts and Analytical Framework

本章界定贯穿全文的四个概念——计数规则、AI损益表、单位接受成果成本与资金地图——并将它们组织为“单位—企业—资本”三层分析框架。

中文

This chapter defines four concepts used throughout, namely the counting rule, the AI P&L, cost per accepted outcome and the map of the money, and organises them into a three-level framework of unit, firm and capital.

ENGLISH

2.1 计数规则

The counting rule

AI收入可分为三个可计数层级：出售智能的实验室、出售AI服务的云厂商、出售成果的应用企业；另有一个不可计数层级，即AI回报内嵌于既有业务、无法单独剥离的企业。使各层级可以相加的前提是一条严格规则：**每一美元终端客户支出只计一次**，计在客户实际交易的层级。实验室从云厂商购买的算力只是同一美元下沉一层；新兴云（neocloud）收入是上层的成本，而非增量需求。

AI revenue falls into three countable tiers, namely labs selling intelligence, clouds selling AI services and applications selling outcomes, plus one uncountable tier: firms whose AI return arrives inside an existing business and cannot be separated from it. What makes the tiers additive is a strict rule: **each end-customer dollar is counted once**, at the tier the customer transacted with. Lab compute bought from a cloud is the same dollar one step down; neocloud revenue is a cost of the tiers above, never incremental demand.

概念界定 DEFINITION

单位接受成果成本 *Cost per accepted outcome*

分子为三项成本：词元（输入与输出）、环境算力（沙箱、检索与运行循环的机器）以及注意力（以满载费率计的人工分钟）；分母为经“裁判”（冻结评测集）接受的成果数，而非产出数量。一个循环产出一百份草稿、交付十二个被接受的成果，其分母即为十二。

The numerator is three costs: tokens (in and out), environment compute (sandboxes, retrieval, the machines the loop runs on) and attention (human minutes at a loaded rate). The denominator is outcomes a referee, a frozen evaluation suite, accepted, not things produced. A loop that yields a hundred drafts and twelve accepted outcomes has a denominator of twelve.

2.2 AI损益表的解剖

The anatomy of an AI P&L

AI产品的损益表出现了软件时代从未有过的科目。收入之下有三项新成本：**智能成本**（按百万词元计价，处于企业无法控制的通缩曲线上）、**证明成本**（评测集、追踪与漂移检查——裁判本身有运行成本）以及**注意力成本**（循环外缘的人工时间，价格曲线触及不到、唯有设计决策能够改变）。其下是对建设者而言按固定计划到来的折旧。净

An AI product's income statement carries lines the software era never had. Below revenue sit three new costs: the **cost of intelligence** (priced per million tokens, on a deflation curve the firm does not control), the **cost of proof** (evaluation suites, traces and drift checks; the referee has a running cost) and the **cost of attention** (human time at the loop's outer edge, untouched by price curves and moved only by design decisions). Beneath them, for the builders,

利润还有“第二引擎”——对客户与供应商持股的估值变动——因此必须先读经营利润。

depreciation arrives on a fixed schedule. Net income also has a second engine, marks on stakes in customers and suppliers, so the operating line is read first.

图 2-1 AI损益表的层级结构
Figure 2-1 *The layered structure of an AI P&L*

收入 Revenue	三个可计数层级加一个不可计数层级；每美元只计一次	Three countable tiers and one uncountable; each dollar counted once
智能成本 Cost of intelligence	按词元计价，处于供应商的通缩曲线上	Tokens priced per million, on the vendor's deflation curve
证明成本 Cost of proof	评测集、追踪、漂移检查的持续开支	Running cost of suites, traces and drift checks
注意力成本 Cost of attention	循环外缘的人工分钟，由设计决定	Human minutes at the loop's edge, moved by design
折旧 Depreciation	建设者按固定时间表承担，与收入无关	Fixed schedule for builders, regardless of revenue
第二引擎 Second engine	持股估值变动；先读经营利润	Marks on stakes; read the operating line first

来源 Source: 根据 Cuofano (2026) 第七章整理 / adapted from Chapter VII.

2.3 四项重写与资金地图

Four rewrites and the map of the money

底本提出四项“重写”。成本线有物理规律：每个词元消耗的算力大致与参数量乘以上下文长度的平方成正比，这解释了为何需求跑赢了五十倍的硬件进步。价格曲线存在悖论：单价下降，而每项任务消耗的词元上升。客户呈重尾：不存在值得据以定价的中位用户。资本离开了资产负债表。资金地图则显示，价值沿技术栈各层**横向**捕获，而风险沿融资链条**纵向**传导，二者几何形状并不重合。

The source proposes four rewrites. The cost line has physics: compute per token scales roughly with parameters times the square of context, which is why demand outran a fifty-fold hardware gain. The price curve has a paradox: unit price falls while tokens per task rise. The customer has a tail: there is no median user worth pricing to. And the capital has left the balance sheet. The map of the money shows that value is captured **horizontally** across the layers of the stack while contagion travels **vertically** through the financing, and the two geometries do not coincide.

表 2-1 本研究的三层分析框架
Table 2-1 *The three-level analytical framework of this study*

层次 Level	核心问题 Core question	主要工具 Main instruments
单位 Unit	每个被接受的成果花费多少、售价多少？ What does each accepted outcome cost, and sell for?	单位接受成果成本；毛利杠杆 Cost per accepted outcome; margin levers
企业 Firm	定价与增长核算如何适应重尾与智能体？ How do pricing and growth accounting adapt to tails and agents?	定价网格；倒转的LTV/CAC Pricing grid; inverted LTV/CAC
资本 Capital	已部署资本需要多少收入，融资结构有多脆弱？ What revenue does deployed capital require, and how fragile is its financing?	折旧覆盖；门槛；酸性测试 Depreciation coverage; hurdle; acid test

第 3 章 · CHAPTER 3

03 研究设计与方法

Research Design and Method

本研究是对底本的结构化分析性阅读：以机制为分析对象，以底本报告的案例与带日期的测量值为证据，并为每一假设设定可证伪条件。

This study is a structured analytical reading of the source volume: mechanisms are the objects of analysis, the cases and dated measurements it reports are the evidence, and every hypothesis carries a falsifier.

中文　　ENGLISH

3.1 资料与分析单位

Material and units of analysis

主要资料为Cuofano（2026）《AI Economics》第一版的全部章节、四个附录及其注释。底本遵循“厂商中立”惯例：正文陈述机制，具体企业名称仅出现在注释中。本研究沿用这一做法，仅在注释可明确对应时才点名。分析单位分为三类：**产品单位**（一个被接受的成果）、**企业**（定价、增长与财务职能）以及**产业资本**（最大建设者的资本开支、折旧与融资结构）。

The primary material is the full first edition of Cuofano (2026), *AI Economics*, including its four appendices and notes. The source follows a vendor-neutral convention: chapters state mechanisms, and named firms appear only in the notes. This study keeps that practice and names firms only where the notes map them unambiguously. There are three units of analysis: the **product unit** (one accepted outcome), the **firm** (its pricing, growth and finance functions) and **industry capital** (the largest builders' capital spending, depreciation and financing structure).

3.2 证据分级

Evidence tiers

表 3-1 底本的证据可信度分级
Table 3-1 *The source's evidence tiers*

证据类型 Evidence type	可信度 Confidence	处理方式 Treatment
监管文件（10-Q/10-K等） Filings (10-Q, 10-K etc.)	高 High	直接引用 Used directly
公司自报年化收入 Company-stated run-rates	中 Medium	未经审计，按“末月×12”理解 Unaudited; read as last month × 12
构造性归因估计 Constructed attributions	低 Low	必须给出区间 Ranges stated
投资者公开言论 Investor statements on stage	标注为言论 Marked as statement	不视为披露 Not treated as disclosure

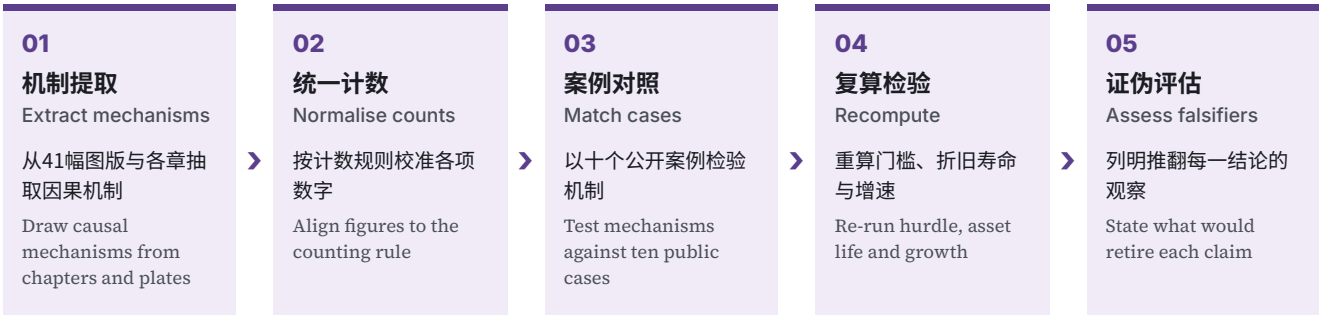
来源 Source: Cuofano (2026), Appendix A.

所有数字均遵循两条计数纪律：**绝不以存量除以流量**，以及**每一美元只计一次**。凡底本未给出精确值者，本研究不补造数字；图表中的复合案例数据均标注为“示意”。

Every number follows two counting disciplines: **never divide a stock by a flow**, and **count each dollar once**. Where the source gives no precise figure, this study

does not supply one; composite-case data in figures are marked illustrative.

图 3-1 本研究的分析流程
Figure 3-1 *The analytical procedure of this study*



3.3 研究假设
Hypotheses

研究假设 HYPOTHESIS

四项可证伪假设 *Four falsifiable hypotheses*

H1：推理毛利是资本账目调和中杠杆最高的单一变量。H2：融资性质从经营现金流转向对手方义务，先于任何信用事件发生。H3：仅凭融资压力，酸性测试综合读数无法超出“脆弱”区间。H4：若AI收入连续四个季度相对折旧保持复合增长，变现之争即告解决。

H1: inference gross margin is the single highest-leverage variable in the capital reconciliation. H2: the shift in funding character, from operating cash flow to counterparty obligations, precedes any credit event. H3: on financing stress alone, the acid-test composite cannot exceed the fragile band. H4: AI revenue compounding against depreciation for four consecutive quarters settles the monetisation argument.

方法说明 METHOD NOTE

测量时点 *Measurement date*

底本的当期测量值均标注为2026年第三季度，作者明言它们是“机制的实例”，将在后续版本中重测。本研究的结论建立在机制之上，而非具体读数之上。

The source's contemporary measurements are dated Q3 2026, and the author states they are instances of mechanisms that will be re-taken in later editions. This study's conclusions rest on the mechanisms, not on the particular readings.

第 4 章 · CHAPTER 4

04 发现一：边际成本回归与毛利的设计化

Finding I: The Return of Marginal Cost and the Designed Margin

在软件时代，毛利率是品类赋予的属性；在AI时代，它是由路由、缓存、自有模型与注意力四个阀门逐一“建造”出来的设计结果。

In software, gross margin was a property of the category; in the AI era it is a design outcome, built valve by valve through routing, caching, an owned model and attention.

中文

ENGLISH

4.1 通缩悖论：单价下降，账单上升

The deflation paradox: unit prices fall, bills rise

底本给出的第一组观察看似矛盾：单位智能的价格在一年内下降约九成，而生产智能的芯片租赁价格却在上涨。机制在于，单价沿供应商曲线下降的同时，智能体为完成每个成果所消耗的词元持续增加，于是单价崩落与人均账单上升可以同时成立。由此推出本研究的第一项发现：以当前价目表编制的预算一个季度后即告过时，唯一耐久的预算形式是**每个被接受成果所消耗的词元数**。

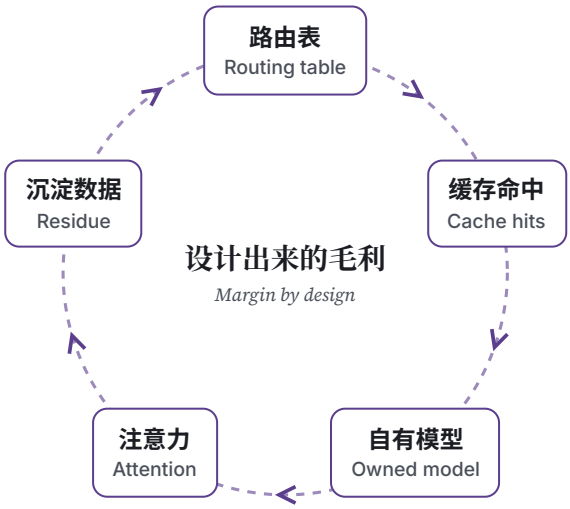
The source's first set of observations looks contradictory: the price of a unit of intelligence fell by roughly ninety percent in a year while the rental price of the chips producing it rose. The mechanism is that unit price falls along the vendor's curve while agents consume ever more tokens per outcome, so a collapsing unit price and a rising bill per user are both true. Hence the first finding: a budget built on today's price table is stale within a quarter, and the only durable form is **tokens per accepted outcome**.

单位接受成果成本之所以是正确的计量器，有三个理由：它与模型无关，更换引擎时分子会如实反映变化；它为注意力定价，而注意力恰是处于企业**自身**通缩曲线上的资源；它是通往损益表的桥梁——乘以数量即为销货成本，与价格比较即为毛利，追踪其趋势即为产品的学习曲线。

Cost per accepted outcome is the right meter for three reasons. It is model-agnostic: swap the engine and the numerator reports the swap honestly. It prices attention, the resource that sits on the firm's **own** deflation curve rather than the vendor's. And it bridges to the P&L: multiplied by volume it is cost of goods, compared with price it is gross margin, and trended it is the product's learning curve.

4.2 毛利的四个阀门
The four valves of margin

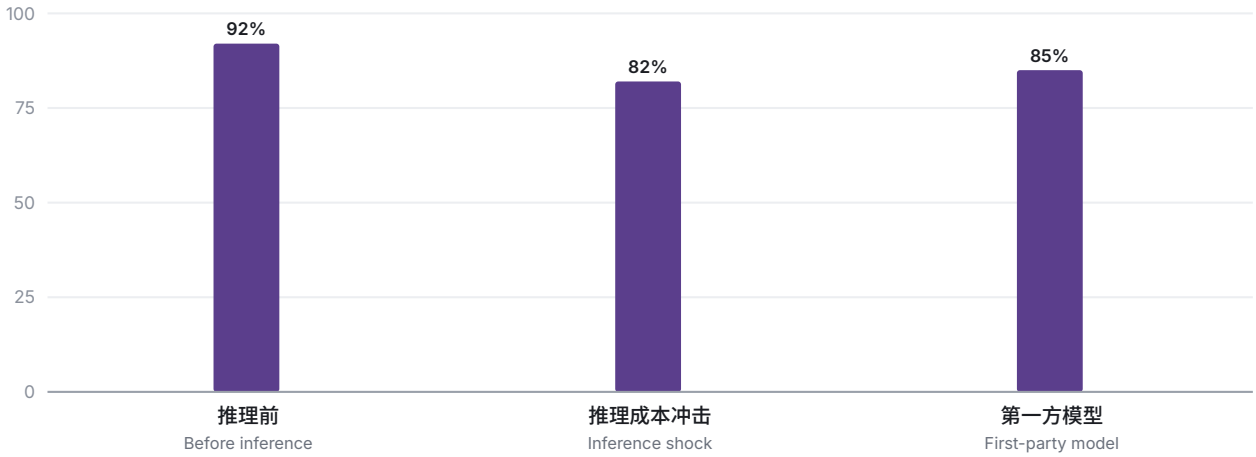
图 4-1 毛利设计的增强循环
Figure 4-1 *The reinforcing loop of margin design*



应用层的案例展示了这一构建过程。底本注释对应的代码编辑器Cursor在租用前沿模型时毛利为负，随后通过三项决策实现微弱的正毛利：把较简单的工作路由给较便宜的模型、缓存重复内容、以及用自身使用数据训练自有模型。设计工具Figma的毛利率因推理成本从92%压缩至82%，又凭第一方模型回升至85%。两次修复都不是规模带来的，而是产品决定了每一单位工作在哪里完成。

Application-layer cases show the building. The code editor the notes identify as Cursor ran negative gross margins while renting frontier models, then reached slight gross-margin profitability through three decisions: routing cheaper work to cheaper models, caching what repeats, and training an owned model on its own usage. The design tool Figma saw a 92 percent margin compress to 82 on inference cost and recover to 85 on a first-party model. Neither recovery came from scale; both came from the product deciding where each unit of work is done.

图 4-2 推理成本对设计工具毛利率的压缩与修复（%）
Figure 4-2 *Compression and recovery of a design tool's gross margin under inference cost (%)*



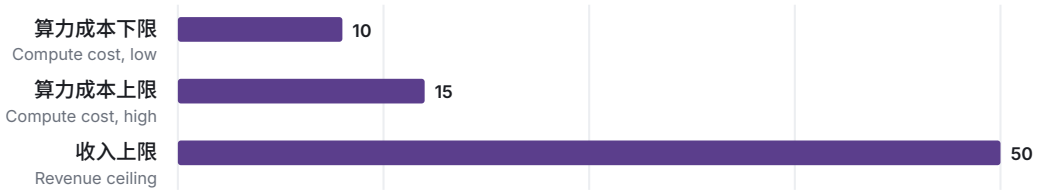
来源 Source: Cuofano (2026), Ch. XI and Appendix D; Figma Q2 2026 print as cited.

4.3 实验室引擎：每兆瓦收入
The lab engine: revenue per megawatt

在模型层，经济单位是兆瓦。实验室以每兆瓦年约1000万至1500万美元的成本购入算力，再以领先的综合费率最高约5000万美元的价格作为认知工作售出：花十，得五十，用差额购买下一个模型。底本特别提醒，这一“兆瓦价差”是最佳容量、最佳价格下的上限；公司层面报告的毛利更接近50%，且两年前仍为负值，因为它涵盖免费层、服务栈、存储与闲置容量等成本。只有后者属于损益表。

At the model layer the economic unit is the megawatt. A lab buys compute at roughly ten to fifteen million dollars per megawatt-year and resells it as cognitive work at up to roughly fifty million at the leading blended rate: spend ten, earn fifty, and buy the next model with the difference. The source warns that this megawatt spread is a ceiling, the best capacity at the best price. The company-level gross margin is nearer fifty percent, and was negative two years ago, because it carries free tiers, the serving stack, storage and idle capacity. Only the latter belongs in a P&L.

图 4-3 每兆瓦年：算力成本与收入上限（百万美元）
Figure 4-3 *Per megawatt-year: compute cost against revenue ceiling ($M)*



来源 Source: Cuofano (2026), Ch. XIV; SemiAnalysis and The Information as cited (Aug 2026). 收入为上限，非平均 / revenue is a ceiling, not an average.

研究发现 FINDING

发现一 *Finding I*

毛利不再是关于行业的事实，而是关于产品决策的事实。四个阀门——路由表、缓存命中率、自有模型产出份额、单位成果注意力——应当被计量，并由财务职能拥有。

Gross margin is no longer a fact about the industry but a fact about the product's decisions. Its four valves, namely the routing table, cache hit rate, owned-model share of outcomes and attention per outcome, should be metered and owned by the finance function.

第 5 章 · CHAPTER 5

05 发现二：席位的断裂与定价网格

Finding II: The Broken Seat and the Pricing Grid

定价从附着于成品的商业决定，转变为决定企业运行何种经济模型的财务工具；增长核算也随“免费用户即成本”而倒转。

Pricing turns from a business decision applied to a finished product into a finance instrument that selects which economics the firm runs; growth accounting inverts once the free user becomes a cost.

中文　　ENGLISH

5.1 席位为何从两端断裂

Why the seat broke from both ends

智能体产品不存在值得据以定价的中位用户：最高十分位的消费量比其余用户高出几个数量级。按中位用户设定的席位价格，恰恰在最值得拥有的客户身上亏损——尾部消耗了公司。另一端，智能体在无人值守的情况下完成工作，按人头计价等于把收入上限锁定在客户的员工数上，而产品恰恰在削减席位所对应的工作。

Agentic products have no median user worth pricing to: the top decile consumes orders of magnitude more than the rest. A seat price set to the median loses money on exactly the customers worth having; the tail consumes the company. At the other end, agents complete work with no one in the chair, so a headcount price caps revenue at the customer's staff count while the product removes the very work the seat represented.

图 5-1 定价网格：可归因性 × 自主性
Figure 5-1 *The pricing grid: attribution × autonomy*

↑ **可归因性** Attribution

人工复核成果 Reviewed outcomes 成果可验证但需人工把关；按成果加复核定价 Verifiable but human-gated; outcome terms with review	**按成果定价** Outcome pricing 系统拥有可验证结果；须以冻结评测集为前提 System owns a verifiable result; needs a frozen suite
按席位定价 Seat pricing 系统辅助人工；仅适用于辅助场景 System assists people; fits assistance only	**按用量定价** Usage pricing 自主行动但难以清晰归因；模型厂商本身即如此收费 Acts alone without clean attribution; how vendors are paid

自主性 Autonomy →

说明 Note: 网格两轴与三种定价取自底本第十二章；左上象限为本研究的补充示意 / axes and three regimes from Ch. XII; top-left quadrant is this study's illustrative extension.

5.2 定价的三条财务规则

Three finance rules for pricing

1. **绝不以席位为智能体价值定价**：它把收入封顶于客户人数。

1. **Never seat-price agentic value**: it caps revenue at the customer's headcount.

2. **为复利定价**：单价随底层成本下降，账户价值随用量上升，应事先向买方展示这条曲线。
3. **为尾部定价**：基础层覆盖中位用户，阶梯层按用量倍数设定，对消费即业务本身的账户采用成果条款。

网格的“梯级”由证明解锁：没有冻结评测集的成果定价只是一份争议日程表。现任者的重新定价正是时代的信号。底本记录，最大的应用软件厂商Salesforce在两年内把其智能体产品从按对话计价改为积分、再改为许可；同时，人类应用使用量持平，而机器协议调用成倍增长。席位出售的是访问权，智能体出售的是工作。

2. **Price the compounding**: unit prices fall as the substrate gets cheaper while account value rises on volume; show the buyer that curve in advance.
3. **Price the tail**: a base that covers the median, tiers in multiples of usage, and outcome terms for accounts whose consumption is the business.

The rungs of the grid are unlocked by proof: outcome pricing without a frozen suite is a dispute schedule. The incumbents' repricing is the era's tell. The source records that the largest application vendor, Salesforce, moved its agent product from a per-conversation price to credits to licences inside two years, while human-application usage stayed flat and machine-protocol calls multiplied. The seat sold access; the agent sells work.

5.3 倒转的增长核算
Growth accounting, inverted

表 5-1 增长核算：软件时代与AI时代的对照
Table 5-1 *Growth accounting: software era versus AI era*

指标 Metric	软件时代 Software era	AI时代 AI era
免费用户 Free user	资产 An asset	成本；免费层只应覆盖到首个被验证成果 A cost; size the free tier to a first verified outcome
生命周期价值 LTV	几乎无销货成本 Near-zero cost of goods	每期带有销货成本；重度用户未必更有价值 Carries COGS per period; heavy users may be worth less
获客成本 CAC	一次付费点击 A paid click	机器给予的引用、智能体使用的接口 A citation the machine grants; an endpoint the agent uses
回本 Payback	订阅期摊销 Amortised over subscription	取决于抵达首个验证成果的时间 Set by time to first verified outcome
增长评分 Growth grade	月活与参与度 MAU and engagement	净收入留存率 Net revenue retention

来源 Source: Cuofano (2026), Ch. XIII and XXIV.

研究发现 FINDING

发现二 *Finding II*

个人层往往亏损，企业循环才是经济本体——领先产品的披露正呈现这种分化。当免费用户成为成本，增长核算必须以被接受的成果及其毛利为基础，楔子由它所找到的循环来买单。

The individual tier is often loss-making and the enterprise loop is the economics; the leading products' disclosures show exactly that split. When the free user is a cost, growth accounting is done on accepted outcomes and their margin, and the wedge is paid for by the loop it finds.

第 6 章 · CHAPTER 6

06 发现三：资本从配置变为义务

Finding III: Capital Turns from Allocation into Obligation

“数千亿资本开支对数百亿AI收入”是一个范畴错误。正确的检验有三个：收入对折旧、资本的回报门槛、以及资金来源的性质。

中文

"Hundreds of billions of capex against tens of billions of AI revenue" is a category error. Three tests work instead: revenue against depreciation, the hurdle the capital requires, and the character of the funding.

ENGLISH

6.1 损益表检验：收入对折旧

The income-statement test: revenue against depreciation

资本开支购买的是能产生五至三十年收入的资产，而收入是十二个月的流量，二者相除会否定历史上任何一次基础设施建设。按计数规则只计一次，AI收入在2025年末首次越过建设者已确认的折旧，此后覆盖倍数一直在一倍左右。然而同期资产购置约为已确认折旧的三倍，这意味着折旧费用将随各批资产到位而升至当前的约三倍。若要维持一倍覆盖，AI收入须在三年内同样翻三倍，即年复合增速约**44%**，而非较宽松读法下的40%。

Capex buys an asset that earns for five to thirty years; revenue is a twelve-month flow, and dividing one by the other condemns every infrastructure build in history. Counted once, AI revenue crossed the builders' recognised depreciation for the first time in late 2025 and has sat near one-times coverage since. Yet purchases ran at roughly three times recognised depreciation, so the charge is heading toward about three times its present level as the vintages land. To hold one-times coverage, AI revenue must also triple over three years, compounding at about **44 percent** a year, not the 40 a looser reading gives.

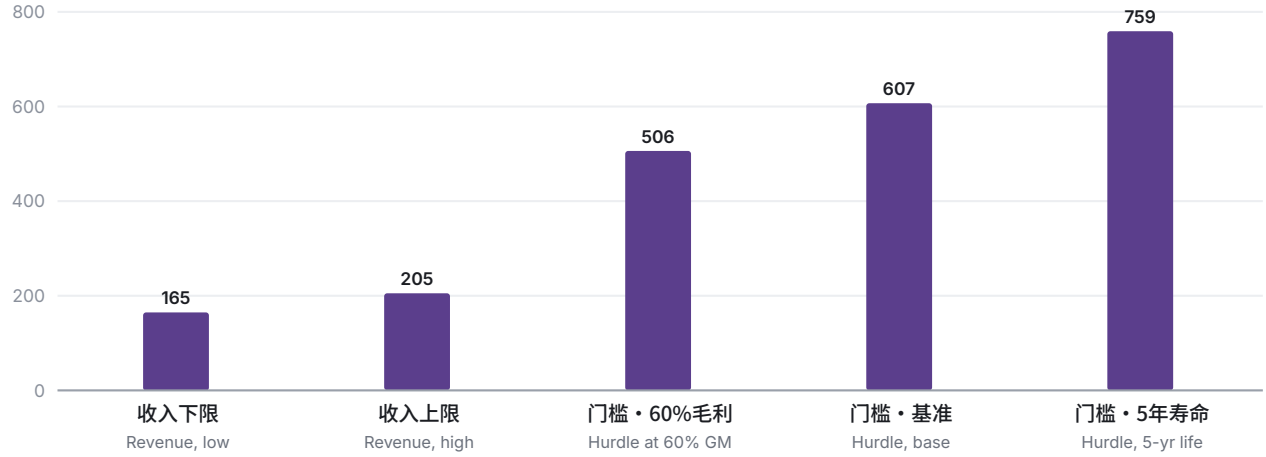
6.2 资本检验：门槛是算术

The capital test: the hurdle is arithmetic

取最大建设者在本周期部署的、可归于AI的约1.25万亿美元资本，要求其获得正常回报：按综合资产寿命计提折旧，再加资本成本，约为资本基数的七分之一加百分之十，约合3000亿美元毛利。按50%毛利率折算，所需年收入约为**6000亿美元**。对照可计数AI收入1650亿至2050亿美元，行业约处于门槛的三分之一处。

Take the roughly $1.25 trillion of AI-attributable capital the largest builders deployed over the cycle and require a normal return: depreciation over a blended asset life plus a cost of capital, about one-seventh of the base plus ten percent, or some $300 billion of gross profit. At a fifty percent gross margin that implies annual revenue of about **$600 billion**. Against countable AI revenue of $165 to $205 billion, the industry stands at roughly a third of the hurdle.

图 6-1 可计数AI收入与回报门槛（十亿美元）
Figure 6-1 *Countable AI revenue against the return hurdle ($ billion)*



来源 Source: Cuofano (2026), Ch. XVI and Appendix B (Q3 2026). 门槛三值为本研究按底本公式复算 / the three hurdle values are this study's recomputation of the source's formula ($1.25T × (1/life + 10%) ÷ GM).

敏感性分析指明了杠杆最高的变量：资产寿命从七年改为五年，门槛上升约四分之一；毛利率从50%提高到60%，门槛下降约六分之一。仅硬件寿命之争就能让所需收入变动约2500亿美元。底本还纠正了一个常见误读：外部的“6000亿美元之问”与本门槛数值接近，并非两条独立路径的相互印证，而是不同年份的数据恰好相遇；按同一方法以当前超过3500亿美元的供应商数据中心年化收入复算，结果接近1.5万亿美元。

Sensitivity names the highest-leverage variables: a five-year asset life instead of seven raises the hurdle by about a quarter; a sixty percent gross margin instead of fifty lowers it by a sixth. The hardware-life dispute alone moves required revenue by about a quarter-trillion dollars. The source also corrects a common misreading: the external "$600B question" sits near this hurdle not because two independent roads converge but because different vintages happen to meet. Re-run on the supplier's current data-centre run-rate, above $350 billion, the same method returns closer to $1.5 trillion.

风险提示 CAUTION

计数的不对称 *The counting asymmetry*

资本按全部建设计入，而收入只在发生交易处计入；最大的单一资本计划属于一家没有外部AI收入、回报体现为广告排序提升的企业。这一不对称使行业看起来比对称口径下离门槛更远。但无论收窄资本口径还是加入AI归因的经营盈余估计，都无法让行业达到门槛。

Capital is counted across the whole build while revenue is admitted only where it transacts; the largest single capital programme belongs to a firm with no external AI revenue, whose return arrives as ad-ranking lift. This asymmetry makes the industry look further from the hurdle than a symmetric measure would. Yet neither narrowing the capital nor adding an estimated AI-attributable operating surplus brings the industry to the hurdle.

6.3 按时钟拆分与融资性质的转变

Splitting by clock and the change in funding character

每一美元资本约六成购买短寿命资产（加速器、服务器、网络，寿命二至六年），约四成购买长寿命资产（机房外壳、变电站、开关设备、土地，寿命十五至四十年）。以五年与二十五年计，年折旧约为基数的13.5%，对应的折

About sixty percent of each capital dollar buys short-lived assets (accelerators, servers, fabric, with lives of two to six years) and about forty percent buys long-lived ones (shells, substations, switchgear, land, fifteen to forty years). At five and twenty-five years, the annual

旧加权寿命约7.3年；若简单算术平均得到十二三年，会把折旧低估近一半。同一美元的两半对同一事件的暴露截然相反：更便宜的模型有利于有电力的机房，却不利于特定载体中的特定一代硬件。

charge is about 13.5 percent of the base, a depreciation-weighted life of about 7.3 years; averaging the lives arithmetically gives twelve or thirteen and understates the charge by nearly half. The two halves of one dollar have opposite exposures: cheaper models help a powered shell and hurt a specific hardware generation in a specific vehicle.

图 6-2　资本美元按资产时钟拆分（%）
Figure 6-2　*The capital dollar split by asset clock (%)*



来源 Source: Cuofano (2026), Ch. XVII. 年折旧率按5年/25年寿命计算 / annual charge at 5- and 25-year lives.

融资检验争议最少、后果最重。两年前，最大建设者的资本开支几乎全部来自经营现金流，增量债务不到资本开支的十分之一；到本周期中段已接近三分之一。五家公司的债券发行占整个投资级市场的六分之一；已签署未起租的租赁达数千亿美元，私募信贷在十八个月内从零增长为可观存量。企业本身没有变，但资金的性质变了：由现金流资助的计划可以随时放缓，由债务、远期租约与私募信贷资助的计划则有对手方、契约与到期日。**资本配置由此变成信用周期。**

The funding test is the least argued and the most consequential. Two years earlier the largest builders funded capex almost entirely from operating cash flow, with incremental debt under a tenth of capital spending; by mid-cycle it was nearly a third. Bonds from five companies became a sixth of the investment-grade market; leases signed but not commenced reached the hundreds of billions; private credit grew from nothing to a large stock in eighteen months. The companies did not change, but the money did: a cash-funded programme can be slowed at will, while one funded by debt, forward leases and private credit has counterparties, covenants and maturities. **Allocation becomes a credit cycle.**

~33%
增量债务占资本开支
incremental debt / capex

~100%
资本开支占经营现金流
capex / operating cash flow

$750B
五大建设者年资本开支
five builders' annual capex

44%
维持折旧覆盖所需增速
growth needed to hold coverage

来源 Source: Cuofano (2026), readings dated Q3 2026. 对比 Comparisons: debt share under 10% two years earlier; capex/OCF ten-year average near 40%.

6.4 酸性测试与“关节”
The acid test and the joints

图 6-3 酸性测试的六个仪表
Figure 6-3 *The six gauges of the acid test*

结构比 Structure ratio	风险加权的表外义务占远期义务比重	Risk-weighted off-book share of forward obligation
速度差 Velocity spread	边际美元离表速度是否快于资本开支增长	Is the marginal dollar leaving the books faster than capex grows
外部依赖 External dependence	外部资本占资本开支比重	External capital as a share of capex
覆盖度 Coverage	远期义务对五年远期现金流	Forward obligation against five years of cash flow
对手方底线 Counterparty floor	次投资级对手方所占份额	Sub-investment-grade share of obligation
接续缺口 Take-out gap	再融资能力相对需求	Refinancing capacity against need

来源 Source: Cuofano (2026), Ch. XIX. 读数 Reading (Q3 2026): stretched, with a 70-point spread between gauges.

酸性测试把六个仪表放在同一刻度上，以便看到它们之间的离散度——综合平均会抹去发现。其核心结论超越了多空两派：把所有融资仪表同时推到压力状态，综合读数最高也只到“脆弱”区间的底部；距离断裂的剩余距离完全取决于需求。融资结构不是泡沫本身，而是泡沫可能借以运行的**放大器**。风险集中在价值地图与信用地图交汇的“关节”处：未评级实验室的算力承诺、次投资级新兴云的租赁、两位数利率的私募信贷，以及与数据中心交叉违约的电力项目。

The acid test scores six gauges on one rail so their spread is visible; a composite would average the finding away. Its headline result beats both camps: push every financing gauge to stress at once and the composite tops out at the bottom of the fragile band, so the remaining distance to a break is entirely the demand question. The financing structures are not the bubble; they are the **amplifier** a bubble would run through. Risk concentrates at the joints where the value map crosses the credit map: an unrated lab's compute commitment, a sub-investment-grade neocloud's lease, double-digit private credit, and power projects cross-defaulted to data centres.

研究发现 FINDING

发现三 *Finding III*

损益表将在到期日之前作答。利润率关口先于接续墙到来：若变现得到证明，证券化市场将成长到足以承接需求；若变现令人失望，接续市场将恰在到期高峰时收缩——两种失败将是同一失败，经由两个渠道相隔约十八个月到来。

The income statement answers before the maturity schedule asks. The margin gate resolves before the take-out wall arrives: if monetisation is demonstrated, the securitisation market grows into the requirement; if it disappoints, the take-out market contracts exactly when maturities peak, and the two failures become one failure arriving through two channels about eighteen months apart.

第 7 章 · CHAPTER 7

07 案例研究与比较分析

Case Studies and Comparative Analysis

本章以底本附录中的复合损益表与十个公开案例为材料，检验前三章的机制能否在具体账目中被观察到。

This chapter uses the source's composite P&L and its ten public cases to test whether the mechanisms of the previous chapters can be seen in concrete ledgers.

中文

ENGLISH

7.1 复合案例：一份损益表的四个季度

Composite case: one P&L over four quarters

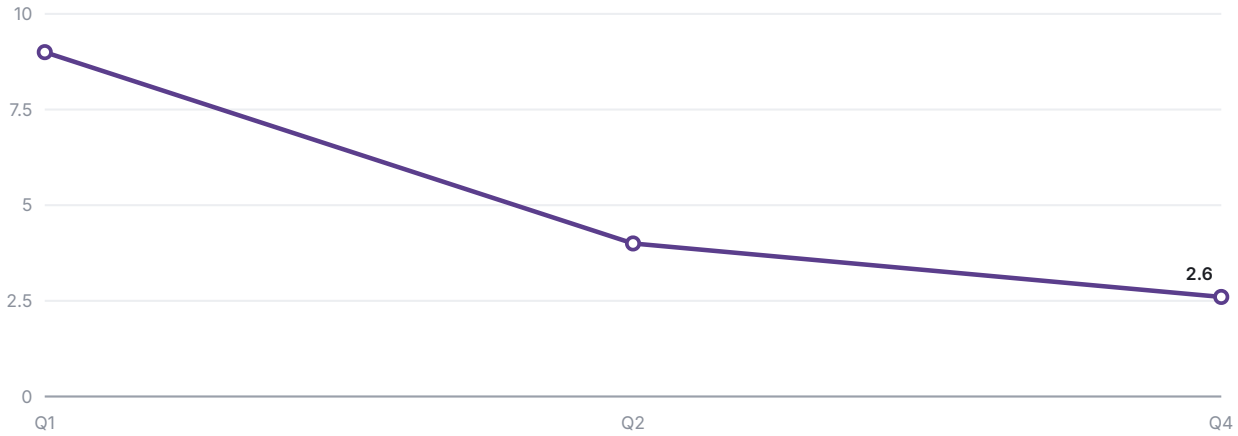
底本附录C构造了一个“没有真实企业、但每一步都真实”的应用AI产品：两千个账户、每席位40美元、毛利率为负15%。第一季度首次计算单位接受成果成本，得出9美元，而席位隐含价格仅为2美元；尾部被识别出来——40个账户消耗了三分之一的推理。第二季度安装路由表：常规成果交给价格仅为十分之一的小模型，并对占请求四成的重复内容进行缓存，单位成本降至4美元，毛利越过零线。

The source's Appendix C builds an applied-AI product that is "no real firm, every step real": two thousand accounts, a forty-dollar seat and a gross margin of minus fifteen percent. In quarter one, cost per accepted outcome is computed for the first time, at nine dollars against a seat-implied price of two, and the tail is named: forty accounts consuming a third of all inference. In quarter two the routing table is installed, sending routine outcomes to a small model at a tenth of the price and caching the forty percent of requests that repeat; unit cost falls to four dollars and gross margin crosses zero.

第三季度在智能体范围内取消席位，改为网格上的三层：辅助用途的基础层、面向智能体用户的按用量层，以及面向40个尾部账户、由评测集验证的按成果条款；尾部账户收入增至三倍，基础层流失下降，因为它不再补贴尾部。第四季度，以产品自身沉淀数据训练的自有模型接管常规层，单位成本降至2.60美元，毛利率达58%；一年后净收入留存率为134%。底本的结论是：品类从未失灵，失灵的是席位。

In quarter three the seat is retired for agentic scope, replaced by three tiers on the grid: a base for assistance, a usage tier for the agent cohort, and suite-verified outcome terms for the forty tail accounts. Revenue per tail account triples and base churn falls, because the base no longer subsidises the tail. In quarter four an owned model trained on the product's own residue takes the routine tier; unit cost reaches $2.60 and gross margin 58 percent, and net revenue retention a year later is 134. The source's verdict: the category was never broken; the seat was.

图 7-1 复合案例：单位接受成果成本的下降路径（美元）
Figure 7-1 *Composite case: the fall in cost per accepted outcome ($)*



示意 Illustrative: 底本附录C的复合案例，非真实企业；第三季度未报告 / composite case from Appendix C, not a real firm; Q3 not reported.

−15%
第零季度毛利率
gross margin, Q0

58%
第四季度毛利率
gross margin, Q4

3×
尾部账户收入增幅
revenue per tail account

134%
一年后净收入留存
NRR one year later

示意 Illustrative: Cuofano (2026), Appendix C composite; no real firm.

7.2 十个公开案例的比较
Comparing the ten public cases

表 7-1 公开案例的分组阅读
Table 7-1 *The public cases, read in groups*

机制 Mechanism	案例（据底本注释） Case (per the notes)	账目上的表现 What the ledger shows
模型层毛利弧线 Model-layer margin arc	Anthropic Anthropic	毛利由深度为负升至约50%；2026年中首次调整后经营盈利 Margin from deeply negative to ~50%; first adjusted operating profit mid-2026
应用层毛利拐点 Application margin turn	Cursor Cursor	2026年4月前实现微弱正毛利；企业账户为正，个人账户亏损 Slight positive margin by April 2026; enterprise positive, individual loss-making
推理成为销货成本 Inference as COGS	Figma Figma	92%→82%→85%，并在席位上加积分计量 92%→82%→85%, adding a credit meter on seats
按成果定价的企业节点 Enterprise junction on outcomes	Palantir Palantir	资本开支低于收入1%，净收入留存高于150% Capex under 1% of revenue; NRR above 150
计算层对照组 Compute-layer pair	CoreWeave、Nebius CoreWeave, Nebius	同一建设，资产负债表相反，融资成本相差数百基点 Same build, opposite balance sheets, priced hundreds of bp apart
供应商充当银行 Supplier as bank	NVIDIA NVIDIA	股权投资、应收账款延长、担保与第三方资本平台 Equity stakes, extended receivables, guarantees, capital platforms
未建设的对照组 The control group	Apple Apple	资本开支低于收入2%，智能以费用化方式租用 Capex under 2% of revenue; intelligence expensed and rented

来源 Source: Cuofano (2026), Appendix D and Notes; figures as of Q3 2026, estimates marked in the source. 底本正文不点名，名称取自注释 / the chapters are vendor-neutral; names come from the notes.

比较揭示了一条共同顺序：资金由内向外移动——先是成本结构，其次是定价，最后才是分发。每一次毛利修复都是产品决策，每一次重新定价都从访问权转向工作，每一处融资风险都位于层级地图上看不见的“关节”。对照组尤其有启发：唯一没有自建的大型企业把智能作为研发费用与租用模型列支，没有经历建设者的任何倒转，其风险体现在合同里而非资产负债表上。它的赌注尚未检验，但说明这些倒转是**可选择的**。

The comparison reveals a shared order: the money moved inside-out, cost structure first, pricing second, distribution last. Every margin recovery was a product decision, every repricing moved from access to work, and every financing risk sat at a joint no layer map shows. The control group is especially telling: the one large firm that did not build expensed its intelligence through research and a rented model, ran none of the builders' inversions, and carried its risk in a contract rather than on its balance sheet. Its bet is untested, but it shows the inversions were **elective**.

案例观察 CASE OBSERVATION

吸收能力阶梯 *The absorption ladder*

财务时钟因公司而异：一家建设者的资本开支占收入三分之一并以现金流自筹，另一家超过一半并进入债券市场、暂停股东回报。同一公司内部，经营时钟与现金时钟也会背离——本季最强的经营吸收恰好叠加在最明显的现金压力之上。

The financial clock is company-specific: one builder ran capex at a third of revenue, self-funded; another ran it above half, entered the bond market and halted returns. Within one company the operating and cash clocks diverge too; the season's strongest operating absorption sat on top of its clearest cash strain.

第 8 章 · CHAPTER 8

08 讨论：启示、局限与风险

Discussion: Implications, Limitations and Risks

建设的账单落在共享瓶颈投入的地方，而支付者大多并非下单者。本章讨论这一判断对三类读者的含义，并检视本研究与底本的局限。

The build's bill lands wherever the bottleneck's inputs are shared, and those who pay it are mostly not those who ordered it. This chapter discusses what that means for three readers and examines the limits of this study and its source.

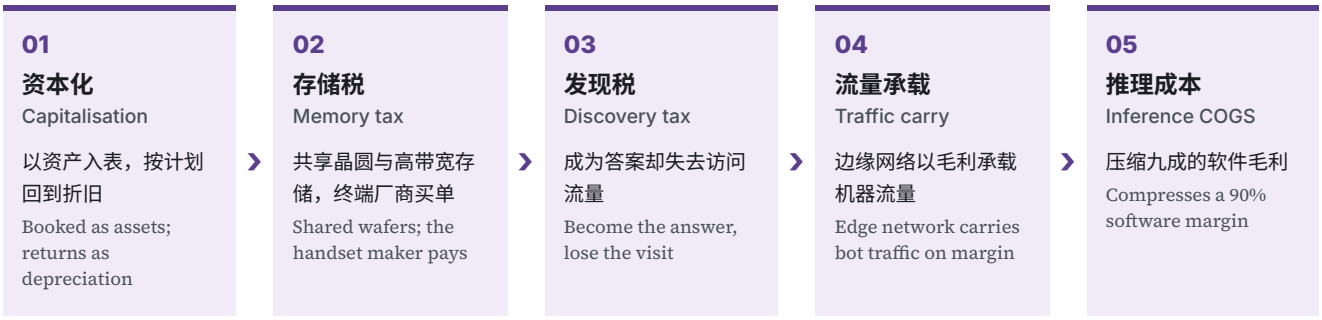
中文

ENGLISH

8.1 账单的五条到达路径

Five paths by which the bill arrives

图 8-1 建设成本的五条归宿路径
Figure 8-1 *The five incidence paths of the build*



来源 Source: Cuofano (2026), Ch. XXVI; Q2 2026 prints of Qualcomm, Reddit, Cloudflare and Figma as cited.

底本把建设的外溢描述为由内向外的五个圆环：物理层（五大建设者一年的资本开支约合二十吉瓦等效建设）、电网（并网排队长达五年）、价格（约每八美元资本开支中有一美元流向三家存储厂商）、劳动与注意力，以及回报。前四个圆环都在等待第五个：只有当智能所完成的工作比智能本身更有价值时，这场建设才会得到偿还。

The source describes the build's spillover as five rings read outward: the physical layer (a year of the five largest builders' capex is about twenty gigawatts of equivalent build); the grid (interconnection queues of five years); prices (roughly one capex dollar in eight goes to three memory makers); labour and attention; and the return. The first four wait on the fifth: the build is repaid only if the work intelligence does is worth more than the intelligence costs.

8.2 对三类读者的启示

Implications for three readers

- **运营者**：以每个被接受成果的词元数编制智能预算；把路由、缓存、自有模型与注意力列为可计量的毛利杠杆。
- **首席财务官**：把定价网格当作财务工具；以收入对折旧替代资本开支对收入；每季度对自身建设与供应商融资运行酸性测试。

- **Operators**: budget intelligence in tokens per accepted outcome; treat routing, caching, the owned model and attention as metered margin levers.
- **Finance chiefs**: treat the pricing grid as a finance instrument; replace capex-against-revenue with revenue-against-depreciation; run the acid test quarterly on the firm's own build and supplier financing.

- **投资者**：价值横向捕获、风险纵向传导，应在两张地图的交点处寻找风险；供应商的创纪录季度意味着更高的门槛，而非更低。

- **Investors**: value is captured horizontally and contagion runs vertically, so look for risk where the two maps cross; a record quarter at the supplier raises the hurdle rather than lowering it.

表 8-1 资金失败的六个家族及其结构性修复
Table 8-1 *Six families of money failure and their structural fixes*

失败家族 Failure family	症状 Symptom	修复 Fix
SaaS毛利假设 SaaS-margin assumption	按80%毛利做计划，实际只有50% Planned on 80%; margin came in at 50	毛利作为设计结果并计量杠杆 Margin as design outcome, levers metered
中位用户定价 Median-user price	尾部吞噬公司 The tail consumes the company	为尾部定价 Price the tail
免费层账单 Free-tier bill	付了词元，用户未转化 Tokens paid, users never converted	免费层仅覆盖到首个验证成果 Wedge sized to a verified outcome
存量除以流量 Stock over flow	以五年资产除以一年收入 Five-year asset over one-year revenue	收入对折旧；义务对远期现金流 Revenue vs depreciation; obligation vs cash flow
表外盲区 Off-balance-sheet blind spot	从损益表读杠杆 Leverage read from the P&L	每季度运行酸性测试 Quarterly acid test
通缩押注 Deflation bet	假设单价下降账单随之下降 Assumed bills fall with unit price	按每成果词元编制预算 Budget on tokens per outcome

来源 Source: Cuofano (2026), Ch. XXVIII.

8.3 研究局限
Limitations

本研究有四项局限：其一，它是对单一作者、单一版本底本的分析性阅读，底本工具缺乏独立复核；其二，许多关键数字是未经审计的自报年化收入或构造性估计；其三，计数规则造成资本宽计、收入窄计的系统性偏差；其四，所有读数截至2026年第三季度，将被替代。

This study has four limitations: it is an analytical reading of one author's single edition, whose instruments lack independent replication; many key figures are unaudited company run-rates or constructed estimates; the counting rule biases the ratio by counting capital broadly and revenue narrowly; and all readings are as of Q3 2026 and will be superseded.

风险提示 CAUTION

主要风险 *Principal risks*

风险并非均匀分布，而是在三类“关节”处集中：未评级实验室—次投资级新兴云—私募信贷—保险与养老金的链条；与数据中心交叉违约的电力项目；以及以五至七年物理时钟投入、却以三至五年信用时钟承保的存储底层。断裂若发生，将自下而上传导：最弱租户先无法融资，随后新兴云重组、基金限制赎回，最后顶层评级调整。

Risk is not evenly spread but concentrated at three joints: the chain from unrated lab to sub-investment-grade neocloud to private credit to insurers and pensions; power projects cross-defaulted to data centres; and a memory floor committing capital on a five-to-seven-year physical clock against demand underwritten on a three-to-five-year credit clock. A break, if it comes, travels upward: the weakest tenant fails to place, then a neocloud restructures, a fund gates, and finally a rating adjusts at the top.

第 9 章 · CHAPTER 9

09 结论与未来研究方向

Conclusions and Future Research Agenda

收入是真实的、增长速度史无前例的，同时又是不足的；本研究的结论是，财务学科必须能够同时持有这三个判断。

The revenue is real, growing faster than anything in recorded history, and insufficient; this study concludes that the finance discipline must be able to hold all three clauses at once.

中文　　ENGLISH

9.1 主要结论

Main conclusions

针对研究问题一，单位层面的答案是：边际成本已经回归，单位接受成果成本是唯一能跨越价格表通缩、如实报告模型替换并为注意力定价的计量器，毛利因此由产品决策建造。针对研究问题二，企业层面的答案是：席位从重尾与智能体两端断裂，定价应沿可归因性与自主性网格向用量与成果迁移，增长核算以被接受成果及其毛利为基础。

On the first research question, the answer at the unit level is that marginal cost has returned, and cost per accepted outcome is the only meter that survives price-table deflation, reports model swaps honestly and prices attention; gross margin is therefore built by product decisions. On the second, the answer at the firm level is that the seat broke from both the heavy tail and the agent, so pricing should migrate along the attribution-and-autonomy grid toward usage and outcomes, and growth accounting should rest on accepted outcomes and their margin.

针对研究问题三，资本层面的答案是：正确的检验是收入对折旧、回报门槛与融资性质，而非资本开支对收入。按Q3 2026读数，收入对折旧约为一倍覆盖，行业约处于6000亿美元门槛的三分之一，融资读数为“紧张”，各仪表间离散70点。针对研究问题四，融资结构是放大器而非泡沫本身；决定结局的是利润率关口能否先于接续墙得到验证。

On the third, the answer at the capital level is that the right tests are revenue against depreciation, the return hurdle and the character of funding, not capex against revenue. On Q3 2026 readings, revenue covers depreciation about one time, the industry stands at roughly a third of a $600 billion hurdle, and the financing reads stretched with a 70-point spread between gauges. On the fourth, the financing structure is an amplifier rather than the bubble itself; the outcome depends on whether the margin gate is passed before the take-out wall arrives.

图 9-1 财务学科的五个动作
Figure 9-1 *The five moves of the discipline*

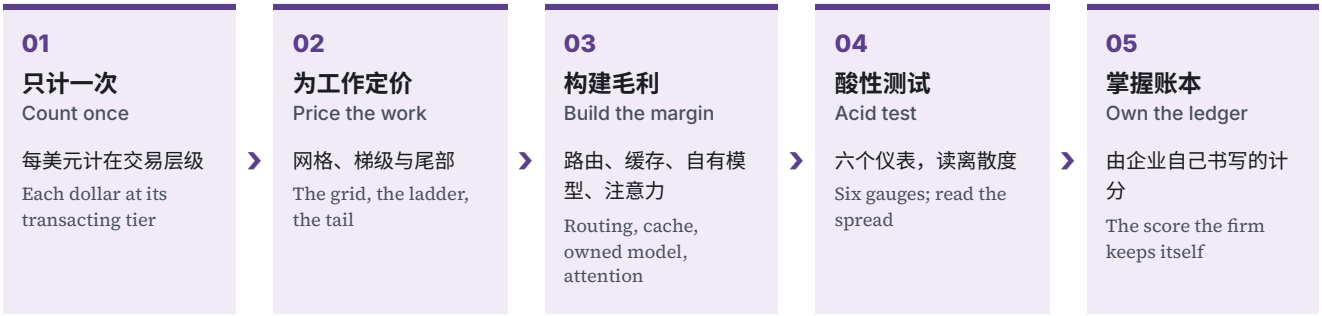


表 9-1 持续跟踪的六个计量器
Table 9-1 *Six meters to track on trend*

计量器 Meter	回答的问题 Question it answers
单位接受成果成本 Cost per accepted outcome	产品的学习曲线 The product's learning curve
推理毛利及其四个杠杆 Inference gross margin and its four levers	毛利是否被设计出来 Is the margin being built
收入对折旧覆盖倍数 Revenue-to-depreciation coverage	损益表能否承担建设 Can the P&L carry the build
资本开支中外部资金占比 External share of capex	资金性质是否改变 Has funding character changed
酸性测试综合值与离散度 Acid-test composite and spread	安装了多少放大器 How much amplification is installed
净收入留存率 Net revenue retention	复利是否发生 Is compounding happening

来源 Source: Cuofano (2026), Ch. XXXIV. 刻意排除：资本开支对收入、年化收入对资本开支、无区间的点估计 / excluded on purpose: capex vs revenue, run-rate vs capex, point estimates without ranges.

9.2 未来研究方向
Future research agenda

1. **对称计量**：建立衡量内嵌于既有业务的AI归因经营盈余（如广告排序提升）的方法，修复资本宽计、收入窄计的偏差。
2. **硬件残值**：利润率关口期间首批AI硬件残值数据到来后，检验折旧寿命假设与五年接续墙的关联。
3. **成果定价的实证**：比较按席位、用量与成果定价企业的毛利与净收入留存，检验定价网格的预测力。
4. **第二桶油**：研究企业自有沉淀数据的定价、确权与许可市场，以及合成数据导致的语料退化。
5. **假设的再检验**：在后续季度用新读数重跑H1至H4及其证伪条件。

1. **Symmetric measurement**: develop methods for the AI-attributable operating surplus embedded in existing businesses, such as ad-ranking lift, to repair the broad-capital, narrow-revenue bias.
2. **Hardware residual value**: once the first residual-value data for AI hardware arrive during the margin gate, test the depreciation-life assumption and its link to the five-year take-out wall.
3. **Evidence on outcome pricing**: compare margins and net revenue retention across seat-, usage- and outcome-priced firms to test the grid's predictive power.
4. **The second barrel**: study pricing, title and licensing markets for firms' own residue, and corpus degradation from generated data.
5. **Re-testing the hypotheses**: re-run H1 to H4 and their falsifiers on later quarters' readings.

> **需求是真实的，而资本要求更多需求；问题在于，损益表能否在到期日发问之前作答。**
>
> *The demand is real and the capital requires more of it, and the question is whether the income statement answers before the maturity schedule asks.*
>
> — Cuofano, *AI Economics*, Epilogue (2026)

参考文献与研究说明 *References & Research Notes*

研究说明 *Note on Method and Sources*

本研究是对Cuofano（2026）《AI Economics》第一版的分析性阅读，以其机制、复合案例与十个公开案例为分析对象。文中数字均为底本报告的、截至2026年第三季度的带日期测量值或作者的估计区间；本研究按底本公式复算的数值及复合案例数据已在图注中标明为“复算”或“示意”。本研究不构成投资建议。

This study is an analytical reading of the first edition of Cuofano (2026), *AI Economics*, taking its mechanisms, composite case and ten public cases as objects of analysis. Figures are the source's dated measurements as of Q3 2026 or its stated estimate ranges; values recomputed from the source's formulas and composite-case data are marked as recomputed or illustrative in the notes. Nothing here is investment advice.

参考文献 *References*

1. Cuofano, G. (2026). *AI Economics — The Discipline of the Money in the AI Era*. The Business Engineer Library. businessengineer.ai
2. Brin, S., & Page, L. (1998). The Anatomy of a Large-Scale Hypertextual Web Search Engine.
3. Shumailov, I., et al. (2024). AI models collapse when trained on recursively generated data. *Nature*.
4. Cahn, D. (2024). AI's $600B Question. Sequoia Capital.
5. a16z. LLMflation: the falling price of a unit of intelligence.
6. Epoch AI (2026). Hyperscaler capital-spending and cash-flow analysis; one-gigawatt data-centre unit economics.
7. Bank of England (2026, July). *Financial Stability Report*.
8. FactSet (2026) on the debt share of hyperscaler capital spending; Morgan Stanley (2026, May) on 2026–27 capital spending.
9. Hyperscaler 10-Q and 10-K filings through Q2 2026 (Alphabet, Amazon, Meta, Microsoft, Oracle).
10. NVIDIA, Q2 FY2027 results (August 2026); SoftBank Group, Q1 FY2026 results (August 2026); Apple, Q3 FY2026 results (July 2026); CoreWeave and Nebius, Q2 2026 results.
11. Salesforce, Agentforce pricing announcements (2025–26); Figma, Q2 2026 results; Palantir, Q2 2026 results; GitHub, usage-based billing documentation (2026).
12. Bloomberg, Forbes and Sacra (2026) on Cursor's run-rate and margin; Bloomberg (2026, August) on Anthropic's run-rate and adjusted operating income.
13. SemiAnalysis (D. Patel) on the Dwarkesh Podcast and The Information (2026, August) on lab gross margins and revenue per megawatt.
14. FourWeekMBA (2023). History of the Internet: From the Internet to Web3.

AI 时代商业工程 · 课题研究系列 · Business Engineering in the AI Era · Research Study Series
No. 04 / 12 · 2026